

Concept DescriptionLesson DescriptionSAS SAS VideoSAS ProgramR Program

Background for an example analysis flag

- In the analysis of clinical trial data, we often summarize data at each analysis visit as shown in the screenshot below.

	Treatment 1				
	Result			Change from Baseline	
	n	Mean	SD	n	Mean
Baseline	XX	XX.X	XX.XX	XX	XX.X
Week 2	XX	XX.X	XX.XX	XX	XX.X
Week 4	XX	XX.X	XX.XX	XX	XX.X
Week 8	XX	XX.X	XX.XX	XX	XX.X
Week 24	XX	XX.X	XX.XX	XX	XX.X

- 'n' in the above table represents the number of subjects with non-missing result at that visit
- Sometimes, there will be multiple records for the same subject and parameter at that analysis visit in the dataset
- For these tables, we need to use only one of those multiple records for analysis
- We need a way to clearly identify which record will be used for analysis in these tables as the existing standard variables (USUBJID, PARAMCD, AVISIT) are not sufficient to identify the record (as there are multiple records with same USUBJID, PARAMCD, AVISIT)
- One approach is to create an additional variable and populate it with a meaningful value to indicate that a record is used for analysis
- The algorithm on which of those multiple records should be prespecified in the statistical analysis plan
- Some common algorithms are:
 - Use the earliest record within the visit
 - Use the latest record within the visit
 - Use the closest record to the target day within the visit
 - Use the average result of all available records within the visit (by creating a new record) etc.

ADaM implementation

- CDISC ADaM standard allows us to handle such scenarios using a set of variables called 'analysis flags'
- The lower-case letters "zz" in the variable name (ANLzzFL) is an index for the zzth variable where "zz" is a padded 2-digit integer [01-99].
- These analysis flags should only be used when the other existing standard variables are not sufficient to identify the record for analysis
- In the above case, as the existing standard variables are not sufficient, we can create an analysis flag variable ANL0IFL and populate it with Y on the records used for analysis